

Property P for two-dimensional spaces with finite angular locus

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Source Question

Mandal–Manna–Paul–Sain define Property P for a Banach space X : there should exist $\varepsilon \in (0, 1)$ such that every bounded linear operator $T \in \mathcal{L}(X)$ preserving Chmielinski ε -approximate Birkhoff–James orthogonality at every point is a scalar multiple of an isometry [1]. Their source page contains the question and definition:

In view of the Blanco-Koldobsky-Turnšek characterization of isometries, it is natural to consider the following more general query:

Given a Banach space $\mathbb{X}$, does there exist an $\epsilon \in (0, 1)$, such that any $T \in \mathcal{L}(\mathbb{X})$ preserving ϵ -orthogonality is an isometry multiplied by a constant?

It is clear that an affirmative answer to the above query will directly lead to a proper refinement of the Blanco-Koldobsky-Turnšek theorem, which is the main motivation behind this study. In order to address the above query, it is convenient to introduce the following definition:

Definition 1.2. Let $\mathbb{X}$ be a Banach space. We say that $\mathbb{X}$ has the *Property P* if there exists an $\epsilon \in (0, 1)$ such that the following two conditions are equivalent:

- (i) $T \in \mathcal{L}(\mathbb{X})$ preserves ϵ -orthogonality at each $x \in \mathbb{X}$.
- (ii) $T \in \mathcal{L}(\mathbb{X})$ is a scalar multiple of an isometry.

Clearly, whenever a Banach space $\mathbb{X}$ possesses the *Property P*, a refinement of the Blanco-Koldobsky-Turnšek theorem is possible for $\mathbb{X}$. On the other hand, if $\mathbb{X}$ does not possess the *Property P*, it follows that the Blanco-Koldobsky-Turnšek theorem can not be improved by considering approximate preservation of orthogonality. Our main aim is to produce concrete examples of both types of Banach spaces by considering certain geometric properties of the concerned space. We first investigate some geometric and analytic properties of finite-dimensional polyhedral Banach spaces, including the smoothness and the number of extreme supporting functionals of each points, related to such preservation. Additionally, we analyze the linear operators satisfying such preservation of orthogonality, in connection with the geometry of the concerned spaces. As an application of this study, we characterize the isometries on some Banach spaces, thus obtaining refinements of the Blanco-Koldobsky-Turnšek theorem on these spaces. Finally, we establish that a two-dimensional Banach space, whose unit sphere is given by a regular $2n$ -gon ($n > 2$), and the sequence space ℓ_∞^n ($n > 2$) do satisfy the *Property P*. On the other hand, we prove that the sequence spaces ℓ_p ($1 \leq p < \infty$) do not possess this property.

We end this section with the following characterization of the approximate Birkhoff-James orthogonality, which will be used repeatedly.

Theorem 1.3. [4, Th. 2.4] *Let $\mathbb{X}$ be a Banach space and let $x, y \in \mathbb{X}$. Let $\epsilon \in [0, 1)$. Then x is ϵ -approximate Birkhoff-James orthogonal to y if and only if there exists $f \in J(x)$ such that $|f(y)| \leq \epsilon \|y\|$.*

2. MAIN RESULTS

We begin this section with the following proposition.

Proposition 2.1. *Let $\mathbb{X}$ be a finite-dimensional polyhedral Banach space. For each $f \in \text{Ext } B_{\mathbb{X}^*}$, consider the set $\mathcal{A}_f = \{x \in \mathbb{X} : J(x) = \{f\}\}$. Then the following results hold true:*

The source proves several positive and negative cases, including regular $2n$ -gon planes for $n > 2$ on the positive side, and ℓ_p for $1 \leq p < \infty$ on the negative side. A separate lane-16 packet removes the regularity hypothesis inside two-dimensional polyhedral spaces. The present packet extends the positive result beyond polyhedral planes.

Definitions and External Tools

For a nonzero vector $x \in X$, write

$$J(x) = \{f \in X^* : \|f\| = 1, f(x) = \|x\|\}$$

for the set of norming functionals. The projective angular locus of a real two-dimensional Banach space X is

$$\mathcal{A}_X = \{[x] \in \mathbb{P}(X) : J(x) \text{ is not a singleton}\}.$$

Thus $[x] \in \mathcal{A}_X$ exactly when the norm is nonsmooth in the direction x . In a polyhedral plane these are precisely the projective vertex lines.

We use the Chmielinski approximate Birkhoff–James relation used in the source:

$$u \perp_B^\varepsilon v \iff \|u + \lambda v\|^2 \geq \|u\|^2 - 2\varepsilon\|u\|\|\lambda v\| \quad \text{for all scalars } \lambda.$$

The source recalls the support-functional characterization: $u \perp_B^\varepsilon v$ if and only if there exists $f \in J(u)$ such that $|f(v)| \leq \varepsilon\|v\|$ [1, Theorem 1.3]. Exact Birkhoff–James orthogonality is denoted by $\perp_B$.

The proof also uses two standard external facts. First, the exact Blanco–Koldobsky–Turnsek theorem says that every linear operator preserving exact Birkhoff–James orthogonality in one direction is a scalar multiple of an isometry [3, 4]. Second, Chmielinski–Stypka prove that if an additive map $T : X \rightarrow Y$ between real normed spaces satisfies the same approximate preservation implication

$$x \perp_B y \implies Tx \perp_B^\varepsilon Ty,$$

then T is linear and bounded and

$$(1 - 8\varepsilon)\|T\|\|x\| \leq \|Tx\| \leq \|T\|\|x\| \quad (x \in X)$$

[2, Theorem 4.4]. We apply this only to already-linear operators; its importance here is the uniform lower bound.

Proof Intuition

The difficulty outside the polyhedral category is that the source’s polyhedral nonsmooth-point lemma is no longer available. The replacement is a compactness argument. If norm-one ε_k -preservers with $\varepsilon_k \downarrow 0$ sent a fixed angular line to smooth lines, then the Chmielinski–Stypka estimate would give a subsequential limit which is an isometry. At the angular source line there are two different supporting functionals, hence two independent exact orthogonality directions. At a smooth image line, however, the unique supporting functional must witness approximate orthogonality for both directions. Passing to the isometric limit forces one nonzero functional to vanish on two independent vectors, impossible.

Consequently, every sufficiently small approximate preserver sends the finite angular locus into itself. Three projective lines in a plane then rigidify a linear map: once their images are prescribed, there is only one projective linear map up to scalar. Thus the possible normalized angular-locus-preserving operators form a finite set. Every nonsimilarity in that finite set has a positive exact-orthogonality defect, and the minimum of these defects supplies the Property P constant.

Lemma 1 (Small preservers respect the finite angular locus). *Let X be a two-dimensional real Banach space with finite angular locus $\mathcal{A}_X$. There is $\alpha_X > 0$ such that whenever $0 < \varepsilon < \alpha_X$ and $S \in \mathcal{L}(X)$ has $\|S\| = 1$ and preserves ε -orthogonality at each point, then*

$$S(\mathcal{A}_X) \subseteq \mathcal{A}_X$$

as projective sets.

Proof. Assume not. Then there are $\varepsilon_k \downarrow 0$, norm-one operators S_k , and projective lines $L_k \in \mathcal{A}_X$ such that each S_k preserves ε_k -orthogonality and $S_k(L_k) \notin \mathcal{A}_X$. Since $\mathcal{A}_X$ is finite, pass to a subsequence for which $L_k = L = [a]$. Choose $a \in L$ with $\|a\| = 1$.

For large k , $\varepsilon_k < 1/8$. The Chmielinski–Stypka estimate gives

$$(1 - 8\varepsilon_k)\|x\| \leq \|S_k x\| \leq \|x\|, \quad x \in X,$$

because $\|S_k\| = 1$. The unit ball of $\mathcal{L}(X)$ is compact in finite dimension, so after another subsequence $S_k \rightarrow S$. Passing to the limit in the displayed inequalities gives $\|Sx\| = \|x\|$ for all $x \in X$. Hence S is a linear isometry, and $S(L) \in \mathcal{A}_X$.

Since L is angular, choose distinct $f_1, f_2 \in J(a)$. Their kernels are distinct one-dimensional subspaces, so we may choose nonzero $y_i \in \ker f_i$ for $i = 1, 2$; then y_1, y_2 are linearly independent and $a \perp_B y_i$.

The line $S_k(L)$ is not angular, so $S_k a$ is a smooth vector. Let h_k be the unique member of $J(S_k a)$. Since S_k preserves ε_k -orthogonality and $a \perp_B y_i$, the support-functional characterization gives

$$|h_k(S_k y_i)| \leq \varepsilon_k \|S_k y_i\|, \quad i = 1, 2.$$

Passing to a subsequence, $h_k \rightarrow h$ in the compact dual unit sphere. Since $S_k a \rightarrow Sa$, we have $h \in J(Sa)$. The displayed inequalities imply

$$h(Sy_1) = h(Sy_2) = 0.$$

But S is invertible and y_1, y_2 are linearly independent, so Sy_1, Sy_2 span X . Hence $h = 0$, contradicting $h \in J(Sa)$. $\square$

Lemma 2 (Finite ray rigidity). *Let X be two-dimensional and let $\mathcal{A}$ be a finite set of at least three one-dimensional subspaces of X . The set of invertible operators which send every line in $\mathcal{A}$ into a line in $\mathcal{A}$ is a finite union of one-dimensional subspaces of $\mathcal{L}(X)$. Consequently, the norm-one such operators form a finite set.*

Proof. For each map $\sigma : \mathcal{A} \rightarrow \mathcal{A}$, set

$$M_\sigma = \{A \in \mathcal{L}(X) : A(L) \subseteq \sigma(L) \text{ for every } L \in \mathcal{A}\}.$$

There are only finitely many σ 's, so it is enough to show that M_σ is at most one-dimensional whenever it contains an invertible operator.

Choose three distinct lines $L_1, L_2, L_3 \in \mathcal{A}$, with L_1, L_2 independent. If M_σ contains an invertible operator, then $\sigma(L_1), \sigma(L_2), \sigma(L_3)$ are distinct and $\sigma(L_1), \sigma(L_2)$ are independent. Pick nonzero $e_i \in L_i$ and nonzero $u_i \in \sigma(L_i)$ for $i = 1, 2$. After scaling e_1, e_2 , write $L_3 = \mathbb{R}(e_1 + e_2)$. Any $A \in M_\sigma$ has

$$Ae_1 = au_1, \quad Ae_2 = bu_2$$

for some scalars a, b . The additional condition $A(L_3) \subseteq \sigma(L_3)$ says that $au_1 + bu_2$ lies in one fixed line distinct from both $\mathbb{R}u_1$ and $\mathbb{R}u_2$. This imposes a fixed homogeneous linear relation between a and b . Thus the possible operators form a linear space of dimension at most one.

Intersecting finitely many one-dimensional operator subspaces with the norm-one sphere gives finitely many points. $\square$

Lemma 3 (Positive gap for nonsimilar angular-line maps). *Let X be a two-dimensional real Banach space for which $\mathcal{A}_X$ is finite and has at least three elements. Among norm-one invertible operators mapping $\mathcal{A}_X$ into $\mathcal{A}_X$, the operators which are not scalar multiples of isometries have a positive lower bound on the ε needed to preserve ε -orthogonality at every point.*

Proof. By Lemma 2, the norm-one invertible operators mapping $\mathcal{A}_X$ into itself form a finite set. Let $\mathcal{F}$ be the subset of those which are not scalar multiples of isometries. If $\mathcal{F} = \emptyset$, there is nothing to prove.

Fix $A \in \mathcal{F}$. By the exact Blanco–Koldobsky–Turnsek theorem, A does not preserve exact Birkhoff–James orthogonality. Hence there are unit vectors $x, y \in X$ such that $x \perp_B y$ but $Ax \not\perp_B Ay$. Therefore no member of $J(Ax)$ annihilates Ay . Since $J(Ax)$ is compact,

$$\eta_A = \min_{g \in J(Ax)} \frac{|g(Ay)|}{\|Ay\|}$$

is strictly positive. If A preserved ε -orthogonality for some $\varepsilon < \eta_A$, the support-functional characterization would produce $g \in J(Ax)$ with $|g(Ay)| \leq \varepsilon\|Ay\|$, a contradiction. Taking the minimum of the finitely many η_A 's over $A \in \mathcal{F}$ gives the asserted positive lower bound. $\square$

Theorem 1 (Finite angular locus criterion). *Let X be a two-dimensional real Banach space. If $\mathcal{A}_X$ is finite and contains at least three projective lines, then X has Property P.*

Proof. Let $\alpha_X > 0$ be supplied by Lemma 1, and let $\delta_X > 0$ be the positive gap from Lemma 3; if the finite set of nonsimilar angular-locus-preserving norm-one maps is empty, take any $\delta_X > 0$. Choose

$$0 < \varepsilon_0 < \min\{\alpha_X, \delta_X, 1/8\}.$$

We claim that ε_0 witnesses Property P.

Let $T \in \mathcal{L}(X)$ preserve ε_0 -orthogonality at each point. If $T = 0$, then T is a scalar multiple of an isometry. Otherwise put $S = \|T\|^{-1}T$. Homogeneity of Chmielinski approximate orthogonality shows that S also preserves ε_0 -orthogonality at each point and $\|S\| = 1$. Since $\varepsilon_0 < 1/8$, the Chmielinski–Stypka estimate makes S bounded below, hence invertible.

By Lemma 1, S maps $\mathcal{A}_X$ into $\mathcal{A}_X$. If S were not a scalar multiple of an isometry, Lemma 3 would force any preservation constant for S to be at least δ_X , contrary to $\varepsilon_0 < \delta_X$. Thus S , and therefore T , is a scalar multiple of an isometry.

Conversely, scalar multiples of isometries preserve exact Birkhoff–James orthogonality, hence preserve ε_0 -orthogonality. Therefore the two conditions in the definition of Property P are equivalent for ε_0 . $\square$

Consequences and Limitations

The theorem strictly extends the previous two-dimensional polyhedral packet. If a two-dimensional polyhedral unit sphere has at least six extreme points, then it has at least three projective vertex lines, and these vertex lines are exactly $\mathcal{A}_X$. The present criterion also applies to non-polyhedral

unit balls with finitely many corners, for instance a centrally symmetric convex disc whose boundary is smooth except at three or more antipodal pairs.

This packet does not settle the full two-dimensional problem. The remaining cases not covered here are:

- finite angular locus of size one or two;
- infinite angular locus;
- the already-negative smooth case, covered separately by the lane-16 uniformly smooth packet.

Verification and Novelty Check

Local run indexes were searched for arXiv:2501.10719, Property P, two-dimensional polyhedral variants, approximate Birkhoff–James preservers, and angular/nonsmooth locus language. Web searches on July 19, 2026 located the source paper’s arXiv and Colloquium Mathematicum records, the 2026 Lebesgue–Bochner continuation by Mohit–Jain, and the 2025 Chmielinski–Stypka paper used here for Theorem 4.4. I found no exact finite-angular-locus criterion or arbitrary non-polyhedral two-dimensional classification.

Human-review recommendation: the main point to check is Lemma 1, especially that the Chmielinski–Stypka relation (4.2) is exactly the same approximate relation as in the source definition. The rest of the proof is a finite projective-rigidity and finite-gap argument.

References

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